SHUBHAM JOSHI

(202) 440-8445 | Email | Portfolio | GitHub | LinkedIn | Hugging Face

APPLIED AI ENGINEER

AI engineer who builds the difficult parts between a model demo and a usable system: governed agents, trained models, evaluation, human review, durable execution, and product interfaces. Built AgentDock, deployed a fine-tuned ModernBERT model, and created a multi-provider generative environment end to end.

SELECTED WORK

AgentDock

TypeScript, Next.js, Prisma, PostgreSQL, Anthropic/OpenAI, MCP · GitHub · Demo

- Built a governed agent control plane that compiles natural-language goals into validated multi-step tool workflows with scoped permissions, approval gates, cost/step limits, and an append-only audit trail.
- Engineered durable execution with a PostgreSQL job queue, worker leases, crash recovery, idempotent external actions, step-cursor resume, SSE events, encrypted BYO keys, and approval-gated Gmail sends.
- Created 61 test files covering policy integrity, tenant ownership, approval semantics, crash recovery, MCP execution, and tool-registration security boundaries.

OpsPilot

Python, PyTorch, ModernBERT, Transformers, FastAPI, PostgreSQL · Model · GitHub

- Fine-tuned and published a ModernBERT support-ticket router on 23,747 leakage-safe examples; raised validation accuracy from 52.3% to 73.1% (weighted-F1 0.713) by redesigning ten overlapping labels into seven operational queues.
- Built confidence-gated routing that auto-routes 62.3% of tickets at 85.5% accuracy and returns uncertain cases for human review; rejected numerically unstable DeBERTa runs instead of reporting invalid gains.

Vibe Curator

TypeScript, WebGL/GLSL, Pixi, OpenAI, Anthropic, ElevenLabs, Gemini Veo · GitHub

- Built an editable generative-media environment that combines generated images, music, video, motion analysis, and parameterized shaders in one persistent, source-aware scene graph.
- Made generated graphics code dependable with static GPU-loop guards, real compilation checks, and compiler-error feedback; shared one WebGL/audio runtime across creation, exploration, and full-screen playback.

CLIENT DELIVERY

ACHP

Analytics Consultant · Johns Hopkins capstone · Alliance of Community Health Plans · 2025-2026

- Led most analytical work on a six-person client engagement; built a reproducible pipeline joining changing CMS schemas into a documented 138,263-row, 208-column master dataset and caught a 12x enrollment overcount before modeling.
- Used enrollment-weighted comparisons, OLS with HC3 robust errors, validated segmentation, county scoring, and plan-exit analysis; showed ACHP held the stronger cost position in 493 of 875 scored counties.
- Presented directly to ACHP leadership on a top-three schoolwide team; the client said the work would change where the organization focused its time.

LEADERSHIP & ADDITIONAL EXPERIENCE

Co-Founder & Vice President · Business Analytics & AI Organization, Johns Hopkins · 2025-2026

- Co-led a 100+ member organization; designed and delivered a three-hour applied-AI workshop and translated agents, RAG, evaluation, safety, and human oversight for mixed technical audiences.

Founder · Digital Marketing & Web Studio · Japan / Remote · 2021-2025

- Shipped websites and campaigns for international clients, owning delivery, client communication, audience research, A/B testing, and performance reporting across time zones.

TECHNICAL SKILLS

AI/ML: PyTorch, TensorFlow, Transformers, ModernBERT, scikit-learn, fine-tuning, NLP, RAG, model evaluation, confidence calibration, structured outputs, human-in-the-loop

Agents & systems: Anthropic/OpenAI APIs, MCP, tool execution, policy gates, memory controls, PostgreSQL queues, workers, idempotency, crash recovery, SSE, testing

Product engineering: Python, TypeScript, JavaScript, React, Next.js, FastAPI, Node.js, PostgreSQL, Prisma, SQLAlchemy, WebGL/GLSL, Git, CI/CD, Railway, Hugging Face

EDUCATION & CREDENTIALS

Johns Hopkins University, Carey Business School · M.S., Business Analytics & Artificial Intelligence · Completed July 2026 · Dean's Scholarship
Nagoya University of Commerce & Business · B.B.A., Global Business Administration · 2025 · Dean's Scholar · Exchange: Monash University
Certifications: Microsoft Azure AI Fundamentals (AI-900) · SAS SQL Essentials · SAS AI Foundations **Languages:** English, Japanese, Nepali (native) · Hindi (fluent)